

BONAVISTA, NL, Canada

WMO: 711780

Lat: 48.670N

Lon: 53.110W

Elev: 26

StdP: 101.02

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.4	-11.6	-17.9	0.8	-12.6	-16.2	0.9	-10.9	25.5	-3.9	23.8	-3.6	11.4	290	1.003

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	24.0	18.9	22.5	18.0	21.0	17.2	19.9	22.9	18.9	21.5	17.9	20.3	7.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	21.5	17.7	12.7	20.4	16.7	12.0	19.5	57.0	23.2	53.7	21.6	50.6	20.3	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	19.3	17.6	-15.3	27.3	2.7	1.0	-17.2	28.0	-18.8	28.6	-20.3	29.2	-22.2	29.9		
(5)			-16.1	21.6	2.8	1.1	-18.1	22.4	-19.8	23.0	-21.3	23.6	-23.4	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-3.5	-4.2	-2.3	1.6	5.4	9.9	15.7	16.5	12.8	8.0	3.7	-0.7
	DBStd	7.85	3.80	4.10	3.61	2.94	3.49	3.88	3.63	2.84	3.15	3.15	3.26	3.57
	HDD10.0	2264	417	398	381	254	151	49	3	0	9	80	189	332
	HDD18.3	4784	676	632	639	503	401	252	94	71	167	321	438	590
	CDD10.0	550	0	0	0	0	8	48	182	201	93	17	1	0
	CDD18.3	29	0	0	0	0	0	1	14	13	2	0	0	0
	CDH23.3	74	0	0	0	0	0	1	43	28	2	0	0	0
	CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0
Wind	WSAvg	8.8	11.0	10.4	9.5	8.4	7.3	7.1	6.8	6.8	8.0	9.2	9.9	10.8
Precipitation	PrecAvg	1068	103	99	93	78	73	76	65	81	90	107	104	105
	PrecMax	1298	177	204	173	145	128	189	164	177	163	181	188	182
	PrecMin	813	37	32	50	21	20	20	27	16	31	51	50	41
	PrecStd	149	35	45	33	31	28	39	30	39	36	36	35	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	8.3	8.6	13.2	19.3	22.7	26.5	25.9	22.8	18.2	14.4	11.1
		MCWB	8.9	7.6	7.3	9.4	13.7	16.9	20.2	20.0	18.6	16.1	13.0	10.3
	2%	DB	5.9	4.8	5.8	10.1	16.2	20.6	24.5	23.9	20.6	16.1	12.4	8.0
		MCWB	5.3	4.2	4.2	7.7	12.0	15.5	18.9	19.0	17.3	14.2	11.5	7.5
	5%	DB	3.2	2.7	3.5	7.7	13.6	18.5	22.9	22.4	19.1	14.3	10.5	5.7
		MCWB	2.6	2.0	2.3	5.6	10.4	14.4	18.1	18.1	16.4	12.6	9.6	5.1
	10%	DB	1.5	1.1	1.9	5.7	11.4	16.4	21.3	21.0	17.7	12.7	8.8	4.0
		MCWB	0.9	0.4	0.8	3.9	8.7	13.2	17.4	17.3	15.2	11.0	7.8	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	7.6	7.6	10.7	14.7	18.2	21.4	21.5	19.8	16.9	13.2	10.6
		MCDB	9.7	8.2	8.4	12.2	17.6	21.6	25.0	24.2	21.8	18.1	13.9	11.1
	2%	WB	5.3	4.4	4.5	8.0	12.7	16.5	20.0	19.9	18.1	14.7	11.5	7.5
		MCDB	5.8	4.7	5.4	9.8	15.2	19.2	23.4	22.9	19.9	15.7	12.2	8.0
	5%	WB	2.7	2.2	2.4	5.9	10.9	14.8	19.0	18.9	16.8	12.9	9.6	5.2
		MCDB	3.1	2.6	3.3	7.5	13.3	17.7	21.7	21.5	18.6	14.1	10.2	5.6
	10%	WB	1.0	0.4	1.1	4.1	8.9	13.3	17.9	17.9	15.5	11.2	7.8	3.4
		MCDB	1.4	0.9	1.8	5.4	11.4	16.1	20.5	20.2	17.3	12.4	8.5	3.9
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.5	5.2	5.4	6.8	7.6	8.0	6.7	6.1	4.9	4.8	4.6
		MCDBR	8.8	8.1	7.5	9.2	11.7	11.9	10.5	9.0	8.1	7.3	7.7	7.7
		MCWBR	9.2	8.1	6.8	7.1	8.0	7.7	6.1	5.3	5.5	6.4	7.7	8.1
	5% WB	MCDBR	8.7	7.9	7.3	9.2	11.0	11.1	9.5	8.3	7.8	7.3	7.6	7.7
		MCWBR	9.1	8.1	7.0	7.5	8.1	7.8	6.2	5.6	6.0	6.8	7.8	8.1
Clear-Sky Solar Irradiance	taub	0.310	0.304	0.336	0.367	0.374	0.401	0.422	0.398	0.371	0.349	0.322	0.309	
	taud	2.311	2.383	2.295	2.330	2.408	2.360	2.348	2.420	2.469	2.471	2.452	2.330	
	Ebn at Noon	740	852	874	881	887	862	836	843	832	790	733	689	
	Edh at Noon	72	84	108	116	112	119	119	106	92	78	63	63	
All-Sky Solar Radiation	RadAvg	0.92	1.70	2.81	4.02	4.78	5.14	5.24	4.41	3.39	2.00	1.11	0.71	
	RadStd	0.08	0.15	0.29	0.40	0.39	0.41	0.49	0.30	0.23	0.22	0.14	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page